



ELECTRONICS DEPARTMENT

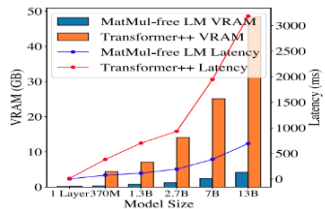
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NOVEL MATMUL-FREE COPROCESSOR ARCHITECTURE FOR LARGE LANGUAGE MODEL

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Abstract

■ This project focuses on designing and implementing hardware acceleration for Token Generation of large language models (LLMs) using a RISC-based pipeline structure. The objective is to address the computational and memory challenges associated with traditional matrix multiplication operations in LLMs by leveraging lightweight MatMul-free architectures.

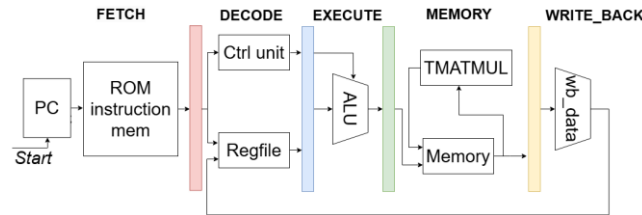


Motivation

■ Eliminating MatMul via ternary quantization and lightweight arithmetic enables efficient, low-power token generation on FPGAs, supporting real-time AI deployment.

Full-precision weight matrix		Ternary Quantization		Input vector		Output
0.94 -0.72 0.15		1 -1 0		2		-3
0.48 0.36 -0.89	→	1 1 -1	×	5	=	4
-0.45 0.23 0.73		-1 0 1		3		1

System overview



■ The system implements a MatMul-free coprocessor architecture based on a 5-stage RISC pipeline (Fetch, Decode, Execute, Memory, Write Back) for token generation in large language models (LLMs). It integrates ternary quantization (weights in $\{-1, 0, +1\}$) to replace costly MatMul operations with simple add/subtract logic. Key features include:

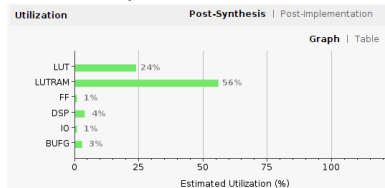
- TMATMUL block for efficient ternary matrix multiplication.
- Custom instruction set for row-wise vector operations, activations, and normalization.

Objectives

- Design and implement a fully MatMul-free coprocessor using ternary quantized weights.
- Develop a custom RISC-style instruction set supporting vectorized operations
- Demonstrate energy and memory efficiency gains on FPGA compared to traditional MatMul-based architectures.

FPGA Implementation

■ FPGA tests show 24% LUT usage, 3.8% DSP usage at 200 MHz. Achieved 87.42% faster than normal multiplication with low power, confirming the efficiency of the MatMul-free



Conclusions

■ This thesis demonstrates the feasibility of deploying large language models without MatMul operations using a ternary-quantized, RISC-style coprocessor. Implemented on an FPGA, the design achieves efficient token generation with significant reductions in latency, power, and memory footprint. This work paves the way for sustainable, high-performance language modeling in resource-constrained environments.

References

- [1] Rui-Jie Zhu et al. "Scalable MatMul-free Language Modeling." 2024.
- [2] Ashish Vaswani et al. "Attention Is All You Need." 2017.
- [3] Digilent. "Genesys 2 Reference Manual." 2025.